Thursday, Jan 06, 2022

Contemporary CH310 Review Questions

(1) Discuss the application of CFSE based on

a. Hydration enthalpy and lattice energies of divalent metal ion(M^{2+})

b. Ionic radii

(2) Explain why π-bonding in tetrahedral complexes is typically much weaker than in octahedral complexes.

(3) (i)Explain why most square planar complexes are favoured with very strong field ligand?
(ii)If substitution process is associative, why may it be difficult to characterize an aqueous ion as labile or inert?

(4) Exchange of H_2O ligand on $[(CO)_3Mn(H_2O)_3]^+$ is much more rapid than on the analogous $[(CO)_3Re(H_2O)_3]^+$. The activation volume (change in volume on formation of the activated complex) is -4.5± 0.4 cm^3mol^{-1}.
(i)Suggest why the Mn complex reacts more rapidly than the analogous Re complex?
(ii)Is the activation volume more consistent with an *A* (or I_a) or a *D* (or I_d) mechanism? Explain.

(5) i) Draw an orbital energy diagram for CO, clearly labelling the σ, π, π* and σ* orbitals. |
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ii) Explain the meaning of the term synergic or dative bonding. Illustrate your answer with an orbital overlap representation of the σ and π symmetry metal-CO interactions. Label the appropriate metal and ligand orbitals used (e.g. d_{xy}, σ_3, etc.).

iii) What effect does strong metal-CO backbonding have on the strength of the C-O bond in the complex? Use your orbital energy diagram from part i) to support your answer.

iv) What experimental/analytical technique(s) would you use to measure the strength of the C-O bond?

(6) Starting from the d-orbital splitting pattern of a hypothetical octahedral metal complex, predict the d-orbital splitting pattern for a complex of trigonal bipyramidal geometry. Remember to assign your x, y and z-axes and ***explain*** your reasoning for any changes to the energies of the respective metal orbitals.

(7) Strong π-acceptor ligands favour late transition metals in low oxidation states.

i) Draw a ***complete*** MO diagram for an octahedral complex having only σ-bonding interactions, such as $[Fe(H_2O)_6]^{2+}$. |

ii) How does replacing the water ligands with CO ligands generate a larger Δ_o? Show the effect on the MO diagram from part i).

(8) i) Define the Jahn-Teller effect.

ii) For which electron configurations would the Jahn-Teller effect be possible?

iii) For which of the electron configurations you proposed in part ii) would the number of unpaired electrons, and thus the effective magnetic moment, be affected if it exhibited Jahn-Teller distortion? Show the resulting orbital splitting diagram resulting from *elongation* of the z-axis.

iv) Identify the d-electron count commonly associated with square planar geometries. Explain why this geometry results in higher stability in the complex.